Executive Summary

TikTok Claims Classification | Exploratory Data Analysis

ISSUE / PROBLEM

TikTok needs to distinguish videos containing factual claims from those expressing opinions to support effective content moderation.

Before developing a machine learning classification model, the data must be explored to understand its structure, quality, and patterns in video engagement.

RESPONSE

Conducted exploratory data analysis (EDA) on 19,382 TikTok video records using Python, Pandas, and NumPy.

Examined dataset structure, missing values, claim classifications, author ban status, and video engagement metrics.

Compared claims and opinions using descriptive statistics, group-level analysis, and engagement ratios to identify meaningful patterns.

IMPACT

The analysis identified important differences between claims and opinions.

Claim videos received approximately 101 times more average views than opinion videos. Claim videos were also associated with a higher proportion of banned authors.

These findings can guide further investigation, data preparation, and feature selection for a future claims classification model.

UNDERSTANDING THE DATA

The dataset contains 19,382 TikTok video records and 12 original variables.

- Claims: 9,608 videos (49.57%)
- Opinions: 9,476 videos (48.89%)
- Missing claim status: 298 videos (1.54%)

Key variables include claim status, author ban status, video views, likes, comments, and shares.

The data contains missing values and highly skewed engagement metrics, which should be considered in future analysis.

```
data['claim_status'].value_counts()
```

```
claim      9608
opinion    9476
Name: claim_status, dtype: int64
```

Note: The counts of each claim status are quite balanced. There are 9,608 claims and 9,476 opinions.

ENGAGEMENT TRENDS

Claim videos showed substantially higher engagement than opinion videos.

- Average views for claims: 501,029
- Average views for opinions: 4,956
- Claim videos received approximately 101 times more average views.

Claims also showed higher likes-per-view, comments-per-view, and shares-per-view ratios across author ban status categories.

These patterns suggest a strong association between claim status and engagement, but do not establish causation.

Claims:

```
Mean view count claims: 501029.4527477102
Median view count claims: 501555.0
```

Opinions:

```
Mean view count opinions: 4956.43224989447
Median view count opinions: 4953.0
```

KEY INSIGHTS

Key Findings:

- Claim and opinion videos are nearly equally represented among labeled records.
- Claim videos receive approximately 101 times more average views than opinion videos.
- 14.98% of claim videos are associated with banned authors, compared with 2.07% of opinion videos.
- Missing values and skewed engagement metrics require attention before predictive modeling.

Recommendation: Investigate these patterns further and assess data quality before developing a claims classification model.

